TIGHTNESS OF THE CYCLE-BASED RELAXATION FOR COMPLETED LENGTH-THREE ALPHA-CYCLES

AYKUT ARSLAN* and META AI

Reviewed by
KIEN LE†

Abstract. Del Pia and Khajavirad [DPK26] introduced the complete edge relaxation for binary polynomial optimization and generalized triangle inequalities for length-three α-cycles. We study the relaxation obtained by adding all switchings of these inequalities to the complete edge relaxation on the completed support of a single length-three α-cycle. If

$$H = \mathrm{cl}(\{e_1, e_2, e_3\})$$

and

$$A = (e_1 \cap e_2) \setminus e_3, \qquad B = (e_2 \cap e_3) \setminus e_1, \qquad C = (e_1 \cap e_3) \setminus e_2,$$

then the cycle-based relaxation is equal to the multilinear polytope if and only if

$$|A| = |B| = |C| = 1.$$

The same characterization remains true for the stronger relaxation obtained by imposing the switched generalized triangle inequalities for every no-private length-three α-cycle contained in H. No restriction is needed on the common intersection or on the private vertices of the three maximal edges. The positive direction is proved by a slice-wise gluing argument, while the negative direction lifts a four-variable parity obstruction and shows that it survives every applicable switched triangle inequality.

1. Introduction

Edited AI

For a hypergraph $H = (V, E)$, the multilinear polytope is the convex hull of the binary vectors satisfying the monomial identities

$$z_e = \prod_{v \in e} z_v, \qquad e \in E.$$

Del Pia and Khajavirad [DPK17, DPK18, DPK26] and Khajavirad [Kha23] studied various relaxations of the multilinear polytope, including the complete edge relaxation $\mathrm{CER}(H)$, obtained by enforcing the exact multilinear polytope separately on each maximal edge. See also [BH02, Cra93, Pad89, SA90] for earlier related works. They proved that this relaxation is tight exactly for α-acyclic hypergraphs and, for α-cycles of length three, introduced generalized triangle inequalities extending the classical triangle inequalities for the Boolean quadric polytope. The graph case goes back to Padberg [Pad89], who showed that the Boolean quadratic polytope coincides with the McCormick relaxation if and only if the underlying graph is acyclic. In 1962, Vorob'ev [Vor62]

2020 *Mathematics Subject Classification.* 90C23, 52B12, 05C65.

*Orchestrating author
†Reviewing author

Edited AI

studied more generally the conditions when pairwise consistent marginals on a set system yield a global extension.

The purpose of this paper is to determine exactly when the generalized triangle inequalities close the remaining gap on the completed support of a single length-three α-cycle. The corresponding characterization question is stated at the end of [DPK26, Section 5.2]. We consider two relaxations. The first adds all switchings of the generalized triangle inequalities for the distinguished private-free reduction of the three maximal edges. The second is stronger: it adds the same inequalities for every no-private length-three α-cycle contained in the completed support.

For an α-cycle of length three, (e_1, e_2, e_3), in a hypergraph $H = (V, E)$, write

$$T = e_1 \cap e_2 \cap e_3,$$

$$A = (e_1 \cap e_2) \setminus e_3, \qquad B = (e_2 \cap e_3) \setminus e_1, \qquad C = (e_1 \cap e_3) \setminus e_2,$$

$$D_i = e_i \setminus (e_j \cup e_k), \qquad \{i, j, k\} = \{1, 2, 3\}.$$

Thus

$$e_1 = T \sqcup A \sqcup C \sqcup D_1, \qquad e_2 = T \sqcup A \sqcup B \sqcup D_2, \qquad e_3 = T \sqcup B \sqcup C \sqcup D_3.$$

For a length-three α-cycle, the three pair-only blocks A, B, C are nonempty, while T and the private blocks D_i may be arbitrary.

Our main theorem is a sharp classification: both cycle-based relaxations are exact if and only if the three pair-only blocks are singletons. The common block and private blocks play no role in the answer. When $T = D_1 = D_2 = D_3 = \emptyset$, the positive case is the ordinary graph triangle.

The proof has two complementary parts, both relying on a probabilistic interpretation of the multilinear polytope originally developed by Pitowsky [Pit91], wherein a point of the multilinear polytope corresponds to a probability distribution on $\{0, 1\}^V$. If one pair-only block contains two vertices, we embed a four-variable parity construction into the three maximal edges. It gives mutually consistent local laws with no global realization. We then show that the resulting point survives every switched generalized triangle inequality coming from every no-private length-three α-cycle in the completed support. Conversely, when A, B, C are singletons, we condition on each state of the common block T. In every slice, the three pairwise marginals on the remaining three variables admit a common extension precisely because the switched generalized triangle inequalities make a natural interval of feasible triple masses nonempty. The slice laws are then glued and the private blocks are restored by conditional sampling.

Statement on AI Use

Human

Meta AI played a substantial role in the research process that led to this manuscript. The work developed through repeated human-directed interaction rather than through a single autonomous run. Meta AI helped explore the question raised by Del Pia and Khajavirad, suggested probabilistic reformulations, identified the four-variable obstruction, and proposed versions of the lifting and gluing arguments used in the final proof.

The mathematical arguments were not accepted on the basis of AI output alone. During the course of the project, several proposed statements and proof steps were found to be incomplete, imprecise, or incorrect. The human authors checked the arguments, corrected gaps, compared the

Human

notation and definitions with the source paper, verified citations, and rewrote substantial parts of the proof. In particular, the probabilistic interpretation of the multilinear polytope, the lifting argument for arbitrary reduced subcycles, and the singleton slice-gluing argument were all subjected to further human review and revision.

Meta AI also assisted with drafting and organizing parts of the exposition. To make this contribution visible, the manuscript uses provenance labels distinguishing material primarily generated by AI, AI-generated material that was subsequently edited or substantially revised by the authors, and material primarily written by the authors. These labels describe the origin and editing history of the text; they are not claims about the mathematical reliability of any particular passage.

The authors determined the research direction, decided which arguments to retain or discard, and took responsibility for resolving mathematical objections that arose during the project. AI-generated claims of correctness were treated as research suggestions rather than as mathematical verification. The final theorem statements, proofs, references, and presentation were reviewed by the human authors, who take responsibility for the contents of the manuscript.

For reproducibility and transparency, links to representative Meta AI conversations used during the development of this work are provided in Appendix A.

2. Preliminaries

Edited AI

2.1. Hypergraphs, completion, and the multilinear polytope. All hypergraphs considered here are finite and have no isolated vertices. For the support of the generating cycle, we take $V = e_1 \cup e_2 \cup e_3$. We identify $z_{\{v\}}$ with the singleton coordinate z_v.

A hypergraph is a pair $H = (V, E)$, where every edge has cardinality at least two. Its completion is

$$\mathrm{cl}(H) = (V, \mathrm{cl}(E)), \qquad \mathrm{cl}(E) = \bigcup_{e \in E} \{f \subseteq e : |f| \geq 2\}.$$

Throughout the main theorem we work with completed hypergraphs.

For a hypergraph $H = (V, E)$, define

$$(H) = \left\{ z \in \{0,1\}^{V \cup E} : z_e = \prod_{v \in e} z_v \text{ for every } e \in E \right\}$$

and

$$\mathrm{MP}(H) = \mathrm{conv}((H)).$$

We also use $z_\emptyset := 1$ as a bookkeeping convention.

For $F \subseteq V$ and $x \in \{0,1\}^V$, write

$$m_F(x) = \prod_{v \in F} x_v,$$

with $m_\emptyset(x) = 1$.

Now we develop a probabilistic framework, analogous to that of Pitowsky [Pit91] in the case of graphs.

Lemma 2.1 (Probabilistic representation)**.** *A point z belongs to $\mathrm{MP}(H)$ if and only if there is a*

Edited AI

probability distribution μ on $\{0,1\}^V$ such that

$$z_v =_\mu [X_v] \quad (v \in V), \qquad z_e =_\mu [m_e(X)] \quad (e \in E).$$

Proof. By definition, $z \in \mathrm{MP}(H)$ if and only if there are coefficients $\lambda_x \geq 0$, $x \in \{0,1\}^V$, with $\sum_x \lambda_x = 1$ and

$$z = \sum_{x \in \{0,1\}^V} \lambda_x L(x),$$

where $L(x)$ is the binary monomial vector associated with x. Defining $\mu(x) = \lambda_x$ gives the required distribution. Conversely, any probability distribution μ yields the convex combination

$$\sum_{x \in \{0,1\}^V} \mu(x) L(x) \in \mathrm{MP}(H).$$

□

Lemma 2.2 (Moment inversion)**.** *Let $e \subseteq V$. A probability distribution on $\{0,1\}^e$ is uniquely determined by the moments*

$$z_F = \left[\prod_{v \in F} X_v\right], \qquad F \subseteq e,$$

where $z_\emptyset = 1$. More precisely, if $x \in \{0,1\}^e$, $O = \{v : x_v = 1\}$, and $Z = e \setminus O$, then

$$(X = x) = \sum_{W \subseteq Z} (-1)^{|W|} z_{O \cup W}.$$

Proof. We have

$$(X = x) = \left[\prod_{v \in O} X_v \prod_{v \in Z} (1 - X_v)\right].$$

Expanding the second product gives the displayed formula. Hence every atom probability is uniquely determined by the subset moments.

Conversely, suppose numbers $(z_F)_{F \subseteq e}$ with $z_\emptyset = 1$ are given, and for each $O \subseteq e$ define

$$p_O = \sum_{F : O \subseteq F \subseteq e} (-1)^{|F \setminus O|} z_F.$$

For every $A \subseteq e$,

$$\begin{aligned} \sum_{O : A \subseteq O \subseteq e} p_O &= \sum_{F : A \subseteq F \subseteq e} z_F \sum_{O : A \subseteq O \subseteq F} (-1)^{|F \setminus O|} \\ &= z_A, \end{aligned}$$

because the inner sum is zero unless $F = A$, when it equals one. Taking $A = \emptyset$ shows that $\sum_O p_O = z_\emptyset = 1$. Therefore, if all p_O are nonnegative, they define a probability distribution on $\{0,1\}^e$ whose moments are exactly $(z_F)_{F \subseteq e}$.

□

2.2. The complete edge relaxation and local laws. Let $H = (V, E)$ be completed and let $E_{\max}$ be its inclusion-wise maximal edges. We define $\mathrm{CER}(H)$ as the set of vectors $z \in \mathbb{R}^{V \cup E}$ such

Edited AI

that for every $e \in E_{\max}$ there is a probability law μ_e on $\{0,1\}^e$ with

$$z_F =_{\mu_e} \left[\prod_{v \in F} X_v\right]$$

for every nonempty $F \subseteq e$, using the singleton coordinates when $|F| = 1$. By Lemma 2.2, this is equivalent to the usual complete-edge inequalities of Del Pia and Khajavirad [DPK17, DPK26].

We call μ_e the local law on e. Since the same moment coordinate is used on overlaps, Lemma 2.2 implies that the local laws associated with a point of $\mathrm{CER}(H)$ are automatically consistent on pairwise intersections.

Combining Lemmas 2.1 and 2.2 gives the dictionary used throughout the paper:

- $z \in \mathrm{CER}(H)$ if and only if z determines a consistent family of local laws on the maximal edges;
- $z \in \mathrm{MP}(H)$ if and only if those local laws admit a common global realization on $\{0,1\}^V$.

2.3. Length-three α-cycles. We use the definition of α-cycle from Del Pia and Khajavirad [DPK26, Definition 2]. For a length-three cycle $C = (e_1, e_2, e_3)$, let

$$S(C) = (e_1 \cap e_2) \cup (e_2 \cap e_3) \cup (e_1 \cap e_3).$$

The defining conditions reduce to

$$e_i \setminus e_j \neq \emptyset \quad (i \neq j)$$

and

$$S(C) \not\subseteq e \qquad \text{for every edge } e \text{ of } H.$$

For the completed support $H = \mathrm{cl}(\{e_1, e_2, e_3\})$, these conditions are equivalent to

$$A \neq \emptyset, \qquad B \neq \emptyset, \qquad C \neq \emptyset.$$

In particular the private blocks D_i are allowed to be nonempty.

A triple $F = (f_1, f_2, f_3)$ has no private vertices if

$$f_i \subseteq f_j \cup f_k, \qquad \{i,j,k\} = \{1,2,3\}.$$

For such a triple, the generalized triangle inequalities are

$$z_{f_1} + z_{f_2} \leq z_{f_1 \cap f_2} + z_{f_3}, \tag{1}$$

$$z_{f_1} + z_{f_3} \leq z_{f_1 \cap f_3} + z_{f_2}, \tag{2}$$

$$z_{f_2} + z_{f_3} \leq z_{f_2 \cap f_3} + z_{f_1}, \tag{3}$$

$$z_{f_1 \cap f_2} + z_{f_1 \cap f_3} + z_{f_2 \cap f_3} \leq z_{f_1 \cap f_2 \cap f_3} + z_{f_1} + z_{f_2} + z_{f_3}. \tag{4}$$

These are the length-three generalized triangle inequalities of [DPK26, Proposition 7].

For $U \subseteq V$, define the switching map on moment coordinates by

$$\phi_U(z_F) = \sum_{W \subseteq F \cap U} (-1)^{|W|} z_{(F \setminus U) \cup W},$$

with $z_\emptyset = 1$. On a deterministic binary point this is precisely the moment obtained after replacing X_v by $1 - X_v$ for $v \in U$. Thus switching preserves $\mathrm{MP}(H)$. This is due to Buchheim and Rinaldi [BR08].

Edited AI

Lemma 2.3. *For every no-private triple $F = (f_1, f_2, f_3)$, inequalities (1)–(4) and all of their switchings are valid for* $\mathrm{MP}(H)$.

Proof. It is enough to check deterministic binary points. The first three inequalities are immediate: for example, if both m_{f_1} and m_{f_2} equal one, then every variable in $f_1 \cup f_2$ equals one, and $f_3 \subseteq f_1 \cup f_2$ because the triple has no private vertices. Hence both terms on the right of (1) equal one. If exactly one of m_{f_1}, m_{f_2} equals one, then $m_{f_1 \cap f_2} = 1$, so the right-hand side is at least one. If both are zero, the inequality is immediate. The other two inequalities are symmetric.

For the fourth inequality, decompose the vertices into the common part and the three pair-only parts of the triple. If a common variable is zero, every term in (4) is zero. Otherwise the inequality reduces, for $x, y, z \in \{0, 1\}$, to

$$x + y + z \leq 1 + xy + xz + yz,$$

which is immediate. Finally, switching maps deterministic moment vectors to deterministic moment vectors, so every switched inequality is also valid. □

For a no-private triple F, let $\mathrm{GT}(F)$ denote the set of points satisfying all switchings of (1)–(4).

For the distinguished three maximal edges, put

$$\bar{e}_i = e_i \setminus D_i = e_i \cap (e_j \cup e_k), \qquad \{i, j, k\} = \{1, 2, 3\},$$

and write

$$\bar{C} = (\bar{e}_1, \bar{e}_2, \bar{e}_3).$$

Because A, B, C are nonempty, $\bar{C}$ is a no-private length-three α-cycle.

We define the distinguished-cycle relaxation

$$\mathcal{R}_0(H) = \mathrm{CER}(H) \cap \mathrm{GT}(\bar{C})$$

and the all-cycle relaxation

$$\mathcal{R}(H) = \mathrm{CER}(H) \cap \bigcap_{F \in \mathcal{C}_0(H)} \mathrm{GT}(F),$$

where $\mathcal{C}_0(H)$ is the set of all no-private length-three α-cycles contained in H. By Lemma 2.3,

$$\mathrm{MP}(H) \subseteq \mathcal{R}(H) \subseteq \mathcal{R}_0(H) \subseteq \mathrm{CER}(H).$$

2.4. Main result.

Theorem 2.4 (Sharp classification)**.** *Let $H = \mathrm{cl}(\{e_1, e_2, e_3\})$ be the completed support of a length-three α-cycle, and define A, B, C, T, D_i as above. Then the following are equivalent:*

1. $|A| = |B| = |C| = 1$;
2. $\mathcal{R}_0(H) = \mathrm{MP}(H)$;
3. $\mathcal{R}(H) = \mathrm{MP}(H)$.

There is no restriction on $|T|$ or on the private blocks D_i. Moreover, if one of A, B, C has size at least two, then

$$\mathcal{R}(H) \setminus \mathrm{MP}(H) \neq \emptyset.$$

The remainder of the paper proves Theorem 2.4.

3. The four-variable obstruction

Edited AI

The support considered here appears in the second part of [DPK26, Example 1]. We use it to construct the obstruction and then prove that the obstruction persists in the larger supports covered by our classification.

The basic obstruction already appears on four variables. Let

$$H_4 = \mathrm{cl}(\{123, 124, 34\}).$$

We label

$$e_1 = 123, \qquad e_2 = 124, \qquad e_3 = 34.$$

On e_1, choose (X_1, X_2, X_3) uniformly from

$$111, \quad 100, \quad 010, \quad 001.$$

On e_2, choose (X_1, X_2, X_4) with the same law. On e_3, let X_3, X_4 be independent fair bits.

Proposition 3.1. *These three local laws are pairwise consistent but have no global realization.*

Proof. The first two laws induce the uniform distribution on (X_1, X_2). The first law makes X_3 fair, the second makes X_4 fair, and these one-dimensional marginals agree with the corresponding marginals of the independent law on (X_3, X_4). Thus the three local laws are pairwise consistent.

The odd-parity law on e_1 implies

$$X_3 = 1 \quad \Longleftrightarrow \quad X_1 = X_2,$$

and the law on e_2 implies

$$X_4 = 1 \quad \Longleftrightarrow \quad X_1 = X_2.$$

Any global realization would therefore satisfy $X_3 = X_4$ almost surely. This contradicts the local law on e_3, under which X_3 and X_4 are independent fair bits. □

By Lemmas 2.1 and 2.2, the common local moment vector belongs to $\mathrm{CER}(H_4)$ but not to $\mathrm{MP}(H_4)$.

4. Negative direction: lifting the obstruction

Assume first that $|A| \geq 2$. Choose distinct vertices $u, v \in A$, together with $b \in B$ and $c \in C$. Fix every vertex outside $\{u, v, b, c\}$ equal to one. On e_1 choose (u, v, c) uniformly from

$$111, \quad 100, \quad 010, \quad 001,$$

on e_2 use the same distribution on (u, v, b), and on e_3 let b, c be independent fair bits. Denote these three local laws by μ_1, μ_2, μ_3, respectively. Denote by z the common moment vector induced by these three local laws.

Proposition 4.1. *The three local laws are pairwise consistent, but they do not admit a global realization. Consequently*

$$z \in \mathrm{CER}(H) \setminus \mathrm{MP}(H).$$

Proof. The first two laws induce the same uniform distribution on (u, v). Moreover, b and c are fair in the corresponding parity laws, so the three laws agree on every pairwise intersection; all remaining variables are fixed equal to one.

Edited AI

Suppose that a global realization existed.

The law μ_1 on e_1 implies

$$X_c = 1 \quad \Longleftrightarrow \quad X_u = X_v,$$

while the law μ_2 on e_2 implies

$$X_b = 1 \quad \Longleftrightarrow \quad X_u = X_v.$$

Thus $X_b = X_c$ almost surely, contradicting the independent fair law on e_3.

Pairwise consistency makes the common moment vector z well defined, and the local-law definition of $\mathrm{CER}(H)$ gives $z \in \mathrm{CER}(H)$. If $z \in \mathrm{MP}(H)$, Lemma 2.1 would give a global probability distribution with these supported moments. Its marginal on each e_i has the same full collection of subset moments as the prescribed local law, hence is equal to that law by Lemma 2.2. This would be a global realization, a contradiction. Therefore $z \notin \mathrm{MP}(H)$. □

We next show that this same point survives every switched generalized triangle inequality coming from every no-private length-three α-cycle in H.

Lemma 4.2 (Structure of a reduced subcycle)**.** *Let $F = (f_1, f_2, f_3)$ be a no-private length-three α-cycle in H. After permuting the three edges, we may assume $f_i \subseteq e_i$ for $i = 1, 2, 3$. Put*

$$P = (f_1 \cap f_2) \setminus f_3, \quad Q = (f_2 \cap f_3) \setminus f_1, \quad R = (f_1 \cap f_3) \setminus f_2, \quad S = f_1 \cap f_2 \cap f_3.$$

Then

$$f_1 = P \sqcup R \sqcup S, \qquad f_2 = P \sqcup Q \sqcup S, \qquad f_3 = Q \sqcup R \sqcup S,$$

and

$$P \subseteq A \cup T, \qquad Q \subseteq B \cup T, \qquad R \subseteq C \cup T, \qquad S \subseteq T.$$

Moreover,

$$P \cap A \neq \emptyset, \qquad Q \cap B \neq \emptyset, \qquad R \cap C \neq \emptyset.$$

Proof. Every edge of H is contained in one of e_1, e_2, e_3. Two of the f_i cannot be contained in the same e_j, because then the union of the three pairwise intersections of F would be contained in e_j, contradicting the α-cycle condition. After relabeling, therefore, $f_i \subseteq e_i$.

Since F has no private vertices, every vertex of $f_1 \cup f_2 \cup f_3$ lies in at least two of the three edges, giving the displayed disjoint decomposition. The containments follow from

$$e_1 \cap e_2 = T \cup A, \qquad e_2 \cap e_3 = T \cup B, \qquad e_1 \cap e_3 = T \cup C.$$

Finally, suppose for example that $P \cap A = \emptyset$. Then $P \subseteq T$, and hence the union of the three pairwise intersections of F satisfies

$$(f_1 \cap f_2) \cup (f_2 \cap f_3) \cup (f_1 \cap f_3) = P \cup Q \cup R \cup S \subseteq T \cup B \cup C \subseteq e_3,$$

contradicting the α-cycle condition. Thus $P \cap A \neq \emptyset$, and the other two claims follow symmetrically. □

Proposition 4.3. *The point z from Proposition 4.1 belongs to $\mathcal{R}(H)$.*

Proof. Fix $F = (f_1, f_2, f_3) \in \mathcal{C}_0(H)$, use the decomposition in Lemma 4.2, and fix a switching set U.

Edited AI

Let

$$X = \prod_{w \in P} \ell_w, \qquad Y = \prod_{w \in Q} \ell_w, \qquad Z = \prod_{w \in R} \ell_w,$$

where $\ell_w = X_w$ for $w \notin U$ and $\ell_w = 1 - X_w$ for $w \in U$. The common block S consists of vertices fixed equal to one. If some vertex of S is switched, then its switched literal is zero, so every term in the four switched generalized triangle inequalities vanishes. We may therefore assume that the product of the switched literals on S equals one.

In this case the switched moments appearing in (1)–(4) are

$$\phi_U(z_{f_1}) =_{\mu_1} [XZ], \qquad \phi_U(z_{f_2}) =_{\mu_2} [XY], \qquad \phi_U(z_{f_3}) =_{\mu_3} [YZ],$$

and

$$\phi_U(z_{f_1 \cap f_2}) = [X], \qquad \phi_U(z_{f_1 \cap f_3}) = [Z], \qquad \phi_U(z_{f_2 \cap f_3}) = [Y],$$

while

$$\phi_U(z_{f_1 \cap f_2 \cap f_3}) = 1.$$

The single-factor expectations may be computed in either incident local law because the local laws agree on overlaps.

Every vertex outside $\{u, v, b, c\}$ is deterministic one. Hence X is either constant, one literal of u or v, or a product of one literal of u and one literal of v. Similarly, Y is constant or one literal of b, and Z is constant or one literal of c.

If one of X, Y, Z is constant, the four inequalities reduce to the Fréchet inequalities for the remaining two variables. For example, if $X = 0$, they reduce to

$$0 \le_{\mu_3} [YZ], \qquad {}_{\mu_3}[YZ] \le_{\mu_3} [Y], \qquad {}_{\mu_3}[YZ] \le_{\mu_3} [Z],$$

$${}_{\mu_3}[Y] +_{\mu_3} [Z] -_{\mu_3} [YZ] \le 1.$$

which hold because (Y, Z) has a genuine joint distribution on e_3. The cases $X = 1$ and the corresponding cases for Y or Z are the same after relabeling.

It remains to consider the case when none of X, Y, Z is constant. We compute

$$p =_{\mu_1} [X], \qquad a =_{\mu_2} [XY], \qquad d =_{\mu_1} [XZ].$$

The expectations involving only Y and Z are computed under μ_3. Consistency of the local laws makes the single-factor expectations agree on overlaps; no joint law of X, Y, Z is assumed.

Since X_b and X_c are independent fair bits under μ_3,

$${}_{\mu_3}[Y] =_{\mu_3} [Z] = \frac{1}{2}, \qquad {}_{\mu_3}[YZ] = \frac{1}{4}.$$

The four switched generalized triangle inequalities become

$$a + d \le p + \frac{1}{4}, \qquad |a - d| \le \frac{1}{4}, \qquad p - a - d \le \frac{1}{4}.$$

If X contains one literal, then $p = 1/2$ and $a = d = 1/4$. If X contains two literals, then X is the indicator of one of the four equally likely values of (u, v), so $p = 1/4$ and $a, d \in \{0, 1/4\}$. In either case the displayed inequalities hold.

Thus z satisfies every switching for every $F \in \mathcal{C}_0(H)$. Together with $z \in \mathrm{CER}(H)$, this proves

Edited AI

$z \in \mathcal{R}(H)$. □

By symmetry, the same construction works if $|B| \geq 2$ or $|C| \geq 2$. Hence if any pair-only block has size at least two, then

$$\mathcal{R}(H) \setminus \mathrm{MP}(H) \neq \emptyset.$$

5. Positive direction: the singleton case

Edited AI

Assume now that

$$A = \{a\}, \qquad B = \{b\}, \qquad C = \{c\}.$$

For $i = 1, 2, 3$, let

$$\bar{e}_i = e_i \setminus D_i.$$

Then

$$\bar{e}_1 = T \cup \{a, c\}, \qquad \bar{e}_2 = T \cup \{a, b\}, \qquad \bar{e}_3 = T \cup \{b, c\}.$$

Lemma 5.1 (Private-block reduction)**.** *Let μ_i be local laws on e_i and let $\bar{\mu}_i$ be their marginals on $\bar{e}_i$. The three laws μ_1, μ_2, μ_3 have a global realization if and only if $\bar{\mu}_1, \bar{\mu}_2, \bar{\mu}_3$ have a global realization on $T \cup \{a, b, c\}$.*

Proof. One direction is immediate by taking a marginal.

Conversely, suppose ν is a global realization of the three reduced laws. For each i, take from μ_i a conditional distribution of the private block D_i given $\bar{e}_i$; on a zero-probability state of $\bar{e}_i$, choose the conditional law arbitrarily. First sample the core variables $T \cup \{a, b, c\}$ according to ν. Then, conditionally on the core, sample the three private blocks independently using these kernels. The resulting law has marginal μ_i on each e_i. □

Let $z \in \mathcal{R}_0(H)$ and let $\bar{\mu}_i$ be the reduced local laws supplied by $\mathrm{CER}(H)$. Fix a state $t \in \{0, 1\}^T$. All masses below are unnormalized slice masses; their total mass in the slice $X_T = t$ is m. Let

$$m = (X_T = t),$$

and define the unnormalized slice moments

$$p_a = (X_T = t, X_a = 1), \quad p_b = (X_T = t, X_b = 1), \quad p_c = (X_T = t, X_c = 1),$$

$$p_{ab} = (X_T = t, X_a = X_b = 1),$$

$$p_{ac} = (X_T = t, X_a = X_c = 1),$$

$$p_{bc} = (X_T = t, X_b = X_c = 1).$$

Each quantity is computed in any reduced local law containing the indicated variables; consistency makes it well defined.

By the Fréchet inequalities [Fré35], for $ij \in \{ab, ac, bc\}$,

$$p_{ij} \geq 0, \qquad p_{ij} \leq p_i, \qquad p_{ij} \leq p_j, \qquad m - p_i - p_j + p_{ij} \geq 0. \tag{5}$$

Let

$$Z_t = \{v \in T : t_v = 0\}.$$

Switch the generalized triangle inequalities for $(\bar{e}_1, \bar{e}_2, \bar{e}_3)$ exactly on Z_t. For every supported

Edited AI

$F \subseteq \{a, b, c\}$ occurring in one of the reduced edges,

$$\phi_{Z_t}(z_{T\cup F}) = \left[\mathbf{1}_{\{X_T=t\}} \prod_{v\in F} X_v\right].$$

Therefore the four switched generalized triangle inequalities become

$$p_{ab} + p_{ac} \le p_a + p_{bc}, \tag{6}$$

$$p_{ab} + p_{bc} \le p_b + p_{ac}, \tag{7}$$

$$p_{ac} + p_{bc} \le p_c + p_{ab}, \tag{8}$$

$$p_a + p_b + p_c \le m + p_{ab} + p_{ac} + p_{bc}. \tag{9}$$

We now extend the three pairwise marginals in this slice to a joint law on (a, b, c). If $m = 0$, then all six quantities $p_a, p_b, p_c, p_{ab}, p_{ac}, p_{bc}$ are zero, and we take the zero measure in this slice. Thus we may proceed with the same formulas for every t. Let r denote a candidate mass assigned to the cell

$$X_T = t, \qquad X_a = X_b = X_c = 1.$$

Once r is chosen, the eight slice cell masses are forced:

(a, b, c)	mass
111	r
110	$p_{ab} - r$
101	$p_{ac} - r$
011	$p_{bc} - r$
100	$p_a - p_{ab} - p_{ac} + r$
010	$p_b - p_{ab} - p_{bc} + r$
001	$p_c - p_{ac} - p_{bc} + r$
000	$m - p_a - p_b - p_c + p_{ab} + p_{ac} + p_{bc} - r.$

Thus all cell masses are nonnegative if and only if

$$L_t \le r \le U_t,$$

where

$$L_t = \max\{0,\ p_{ab} + p_{ac} - p_a,\ p_{ab} + p_{bc} - p_b,\ p_{ac} + p_{bc} - p_c\}$$

and

$$U_t = \min\{p_{ab}, p_{ac}, p_{bc},\ m - p_a - p_b - p_c + p_{ab} + p_{ac} + p_{bc}\}.$$

Lemma 5.2 (Slice feasibility)**.** *For every $t \in \{0, 1\}^T$, the interval $[L_t, U_t]$ is nonempty.*

Proof. The lower bound 0 is at most p_{ab}, p_{ac}, p_{bc} by (5), and it is at most the final upper bound by (9).

Consider

$$\ell_a = p_{ab} + p_{ac} - p_a.$$

Since $p_{ab} \le p_a$ and $p_{ac} \le p_a$,

$$\ell_a \le p_{ac}, \qquad \ell_a \le p_{ab}.$$

Edited AI

Inequality (6) gives $\ell_a \le p_{bc}$. Finally,

$$\begin{aligned} & m - p_a - p_b - p_c + p_{ab} + p_{ac} + p_{bc} - \ell_a \\ &= m - p_b - p_c + p_{bc} \ge 0 \end{aligned}$$

by (5) for the pair (b, c). The same argument after cyclically permuting a, b, c handles the other two nonzero lower bounds. Hence every lower bound defining L_t is at most every upper bound defining U_t. □

For each t, choose $r_t \in [L_t, U_t]$ and use the table above to define a nonnegative measure in the slice $X_T = t$. Its total mass is m. Summing over all t therefore gives a probability distribution ν on $T \cup \{a, b, c\}$. By construction, its marginals on

$$T \cup \{a, c\}, \qquad T \cup \{a, b\}, \qquad T \cup \{b, c\}$$

are exactly $\bar{\mu}_1, \bar{\mu}_2, \bar{\mu}_3$.

Proposition 5.3. *If $|A| = |B| = |C| = 1$, then*

$$\mathcal{R}_0(H) = \mathrm{MP}(H).$$

Consequently $\mathcal{R}(H) = \mathrm{MP}(H)$ as well.

Proof. Let $z \in \mathcal{R}_0(H)$. The construction above gives a global realization of the reduced local laws. Lemma 5.1 restores the private blocks and yields a global distribution whose supported moment vector is z. Lemma 2.1 therefore gives $z \in \mathrm{MP}(H)$. Thus $\mathcal{R}_0(H) \subseteq \mathrm{MP}(H)$. The reverse inclusion follows from Lemma 2.3 and the validity of the complete edge relaxation. Since

$$\mathrm{MP}(H) \subseteq \mathcal{R}(H) \subseteq \mathcal{R}_0(H),$$

the equality $\mathcal{R}(H) = \mathrm{MP}(H)$ follows as well. □

6. Proof of the main theorem

Edited AI

Proof of Theorem 2.4. If $|A| = |B| = |C| = 1$, Proposition 5.3 gives

$$\mathcal{R}_0(H) = \mathcal{R}(H) = \mathrm{MP}(H).$$

Conversely, suppose one of A, B, C has size at least two. By symmetry assume $|A| \ge 2$. Proposition 4.1 constructs a point $z \in \mathrm{CER}(H) \setminus \mathrm{MP}(H)$, and Proposition 4.3 shows that the same point belongs to $\mathcal{R}(H)$. Hence

$$\mathcal{R}(H) \setminus \mathrm{MP}(H) \neq \emptyset.$$

Since $\mathcal{R}(H) \subseteq \mathcal{R}_0(H)$, neither cycle-based relaxation is equal to $\mathrm{MP}(H)$. This proves all equivalences. □

7. Conclusion

Human

For the completed support of a length-three α-cycle, the exactness of generalized triangle inequalities is controlled entirely by the three pair-only intersections. The complete edge relaxation together with all switchings of the distinguished generalized triangle inequalities is exact precisely when each pair-only block consists of a single vertex. Adding the corresponding inequalities from every other no-private length-three α-cycle in the completion does not enlarge the class of exact

Human

instances.

The proof separates the roles of the different parts of the Venn decomposition. Private blocks can be removed and restored by conditional kernels, and the common block can be handled slice by slice. The obstruction appears only when a pair-only block contains two vertices, where a four-variable parity construction creates locally compatible edge laws that cannot be globally realized while remaining invisible to all of the switched generalized triangle inequalities considered here.

A natural next question is whether analogous sharp classifications can be obtained for longer α-cycles or for stronger families of inequalities beyond the length-three generalized triangle inequalities.

Appendix A. AI Chats

The following Meta AI conversations document representative stages of the research process leading to this manuscript.

- `https://meta.ai/share/c/xrb8TU157j`: Research conversation used during the development and checking of the alpha-cycle proof.
- `https://meta.ai/share/c/8DSNlAr2E3`: Reorganization of the manuscript into an expert-reader format, including proof ordering, exposition, and AI/human attribution conventions.

References

[BH02] Endre Boros and Peter L. Hammer. Pseudo-Boolean optimization. *Discrete Applied Mathematics*, 123(1):155–225, 2002.

[BR08] Christoph Buchheim and Giovanni Rinaldi. Efficient reduction of polynomial zero-one optimization to the quadratic case. *SIAM Journal on Optimization*, 18(4):1398–1413, 2008.

[Cra93] Yves Crama. Concave extensions for non-linear 0–1 maximization problems. *Mathematical Programming*, 61:53–60, 1993.

[DPK17] Alberto Del Pia and Aida Khajavirad. A polyhedral study of binary polynomial programs. *Mathematics of Operations Research*, 42(2):389–410, 2017.

[DPK18] Alberto Del Pia and Aida Khajavirad. The multilinear polytope for acyclic hypergraphs. *SIAM Journal on Optimization*, 28(2):1049–1076, 2018.

[DPK26] Alberto Del Pia and Aida Khajavirad. The complete edge relaxation for binary polynomial optimization. *arXiv preprint arXiv:2507.12831*, 2026. Version v2, 10 June 2026.

[Fré35] Maurice Fréchet. Généralisation du théorème des probabilités totales. *Fundamenta Mathematicae*, 25:379–387, 1935.

[Kha23] Aida Khajavirad. On the strength of recursive McCormick relaxations for binary polynomial optimization. *Operations Research Letters*, 51(2):146–152, 2023.

[Pad89] Manfred Padberg. The Boolean quadric polytope: some characteristics, facets and relatives. *Mathematical Programming*, 45(1–3):139–172, 1989.

[Pit91] Itamar Pitowsky. Correlation polytopes: their geometry and complexity. *Mathematical Programming*, 50(1-3):395–414, 1991.

[SA90] Hanif D. Sherali and Warren P. Adams. A hierarchy of relaxations between the continuous and convex hull representations for zero-one programming problems. *SIAM Journal on Discrete Mathematics*, 3(3):411–430, 1990.

[Vor62] N. N. Vorob'ev. Consistent families of measures and their extensions. *Theory of Probability and its Applications*, 7(2):147–163, 1962.